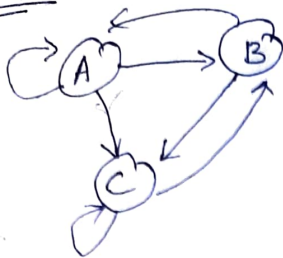


Page Rank Algorithm.

Example 1



Create Transition matrix

$$(M) \begin{array}{c|ccc} & A & B & C \\ \hline A & \swarrow 1/3 & \swarrow 1/2 & \rightarrow 0 \\ B & 1/3 & 0 & 1/2 \\ C & 1/3 & 1/2 & 1/2 \end{array}$$

$$\underline{\underline{E}} \quad \underline{\underline{1}} \quad \underline{\underline{1}} \quad \underline{\underline{1}} \quad (\text{verify}).$$

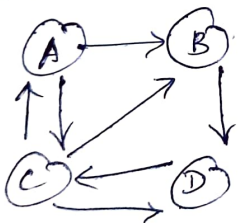
Initial $R_0 = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}$ equal weights for all nodes.

Iteration 1 $R_1 = M R_0 = \begin{bmatrix} 1/3 & 1/2 & 0 \\ 1/3 & 0 & 1/2 \\ 1/3 & 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}$

Iteration 2 $R_2 = M R_1$. (converge for fixed iteration or when 2 iteration values are same).

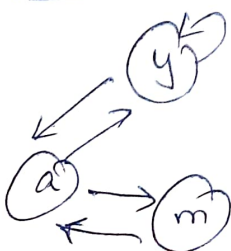
In general, $R_{i+1} = M * R_i$

Example 2



	A	B	C	D
A	0	0	1/3	0
B	1/2	0	1/3	0
C	1/2	0	0	1
D	0	1	1/3	0

Example 3



	y	a	m
y	1/2	1/2	0
a	1/2	0	1
m	0	1/2	0

Soln
Page rank order.
a, y, m.

Example 3 solved

$$\text{Iteration 1} \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0.5 & 0 & 1 \\ 0 & 0.5 & 0 \end{pmatrix} \begin{pmatrix} 0.33 \\ 0.33 \\ 0.33 \end{pmatrix} = \begin{pmatrix} 0.33 \\ 0.495 \\ 0.165 \end{pmatrix}$$

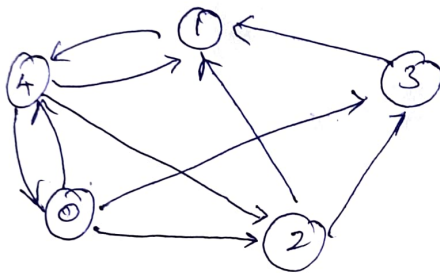
$$\text{Iteration 2} \begin{pmatrix} -do- \end{pmatrix} \begin{pmatrix} 0.33 \\ 0.495 \\ 0.165 \end{pmatrix} = \begin{pmatrix} 0.4125 \\ 0.33 \\ 0.2475 \end{pmatrix}$$

$$\text{Iteration 3} \begin{pmatrix} -do- \end{pmatrix} \begin{pmatrix} 0.4125 \\ 0.33 \\ 0.2475 \end{pmatrix} = \begin{pmatrix} 0.371 \\ 0.45 \\ 0.165 \end{pmatrix} \Rightarrow \text{order } a, y, m.$$

Example 4 (with damping factor d)

default $d = 0.85$

$$R_{i+1} = (M * R_i) d + (1-d) \left(\frac{1}{n} \right).$$



	0	1	2	3	4
0	0	0	0	0	1/3
1	0	0	1/2	1	1/3
2	1/3	0	0	0	1/3
3	1/3	0	1/2	0	0
4	1/3	1	0	0	0

Iteration 1

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0.33 \\ 0 & 0 & 0.5 & 1 & 0.33 \\ 0.33 & 0 & 0 & 0 & 0.33 \\ 0.33 & 0 & 0.5 & 0 & 0 \\ 0.33 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1/5 \\ 1/5 \\ 1/5 \\ 1/5 \\ 1/5 \end{bmatrix} \cdot 0.85 + (1-0.85) \begin{bmatrix} 1/5 \\ 1/5 \\ 1/5 \\ 1/5 \\ 1/5 \end{bmatrix}$$

$$= \begin{bmatrix} 0.086 \\ 0.341 \\ 0.142 \\ 0.171 \\ 0.256 \end{bmatrix}$$

Iteration 2

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0.33 \\ 0 & 0 & 0.5 & 1 & 0.33 \\ 0.33 & 0 & 0 & 0 & 0.33 \\ 0.33 & 0 & 0.5 & 0 & 0 \\ 0.33 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0.086 \\ 0.341 \\ 0.142 \\ 0.171 \\ 0.256 \end{bmatrix} \cdot 0.85 + (0.15) \begin{bmatrix} 0.2 \\ 0.2 \\ 0.2 \\ 0.2 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 0.101 \\ 0.307 \\ 0.125 \\ 0.114 \\ 0.344 \end{bmatrix}$$

R_1 - Page 4, R_2 - Page 1, R_3 - Page 2, R_4 - Page 3, R_5 - Page 0

Iteration 1

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0.33 \\ 0 & 0 & 0.5 & 1 & 0.33 \\ 0.33 & 0 & 0 & 0 & 0.33 \\ 0.33 & 0 & 0.5 & 0 & 0 \\ 0.33 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0.2 \\ 0.2 \\ 0.2 \\ 0.2 \\ 0.2 \end{bmatrix} \cdot 0.85 + (1-0.85) \cdot \begin{bmatrix} 0.2 \\ 0.2 \\ 0.2 \\ 0.2 \\ 0.2 \end{bmatrix}$$

$$\begin{bmatrix} 0.066 \\ 0.366 \\ 0.132 \\ 0.166 \\ 0.266 \end{bmatrix} \cdot 0.85 + 0.15 \begin{bmatrix} 0.2 \\ 0.2 \\ 0.2 \\ 0.2 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 0.056 \\ 0.311 \\ 0.112 \\ 0.141 \\ 0.226 \end{bmatrix} + \begin{bmatrix} 0.03 \\ 0.03 \\ 0.03 \\ 0.03 \\ 0.03 \end{bmatrix}$$

$$= \begin{bmatrix} 0.086 \\ 0.341 \\ 0.142 \\ 0.171 \\ 0.256 \end{bmatrix} \quad [2M]$$

Iteration 2

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0.33 \\ 0 & 0 & 0.5 & 1 & 0.33 \\ 0.33 & 0 & 0 & 0 & 0.33 \\ 0.33 & 0 & 0.5 & 0 & 0 \\ 0.33 & 1 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0.2 \\ 0.2 \\ 0.2 \\ 0.2 \\ 0.2 \end{bmatrix} \cdot 0.85 + 0.15 \begin{bmatrix} 0.2 \\ 0.2 \\ 0.2 \\ 0.2 \\ 0.2 \end{bmatrix}$$

$$= \begin{bmatrix} 0.084 \\ 0.326 \\ 0.112 \\ 0.099 \\ 0.369 \end{bmatrix} \cdot 0.85 + \begin{bmatrix} 0.03 \\ 0.03 \\ 0.03 \\ 0.03 \\ 0.03 \end{bmatrix}$$

$$= \begin{bmatrix} 0.071 \\ 0.277 \\ 0.095 \\ 0.084 \\ 0.314 \end{bmatrix} + \begin{bmatrix} 0.03 \\ 0.03 \\ 0.03 \\ 0.03 \\ 0.03 \end{bmatrix}$$

$$= \begin{bmatrix} 0.101 \\ 0.307 \\ 0.125 \\ 0.114 \\ 0.344 \end{bmatrix}$$

R1 - Page 4
R2 - Page 1
R3 - Page 2
R4 - Page 3
R5 - Page 0

R1 - Page 4, R2 - Page 1, R3 - Page 2, R4 - Page 3, R5 - Page 0